Limitations of Q-Learning

- **Scalability**: Struggles with large state or action spaces due to the size of the Q-Table.
- **Exploration vs. Exploitation**: Balancing exploration of new actions and exploitation of known rewards can be challenging.
- **Requires Discrete Spaces**: Traditional Q-Learning is not suited for continuous state or action spaces.

Q-Learning forms the foundation of many advanced reinforcement learning algorithms. By understanding its mechanics and practical applications, developers can design intelligent agents capable of learning and adapting to diverse environments.

Real-World Example: Building an AI for a Simple Game

Reinforcement Learning (RL) is a powerful paradigm that finds significant applications in gaming. By allowing agents to learn optimal strategies through repeated interactions with the game environment, RL opens up possibilities for creating intelligent gameplaying systems. One of the simplest yet effective examples of RL in action is building an AI for Tic-Tac-Toe. This classic game offers a manageable environment for demonstrating the principles of state representation, action selection, and reward-based learning.